Impression updating: The role of change detection and recollection in associative memory for face-adjective pairs

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Abstract

Age differences in associative memory include differences in the updating of learned associations with new information. The present study investigated younger and older adults' memory for changed associations between faces and adjectives. Participants were presented with pictures of faces paired with adjectives that described positively- or negatively-valenced personal attributes (e.g., honest, cowardly), and were instructed to study these pairs and form an impression of each person based on the paired adjective. Immediately after study, participants provided an initial impression rating of each person on a 6-point scale from very negative to very positive. A second encoding phase consisted of face-adjective pairs that included new faces and new adjectives, old faces with the same adjectives, and old faces with different adjectives. Participants were instructed again to form an impression of each person, updating the previous impression if the adjective had changed, and indicate whether the face-adjective pair was the same, different, or new. In the test phase, participants viewed each face and attempted to recall the adjective from the second encoding phase. They also indicated whether they remembered a change in the adjective across study phases, and provided a final impression rating on the 6-point scale. Results indicate both older and younger adults struggle with the adjective cued recall task, yet memory performance improved when participants were asked to complete a change detection task. Age differences were observed between younger and older adults' change detection results, such that older adults' change detection and impression ratings depended on the valence of the updated adjective: when the second adjective was negative, older adults detected the adjective change as often as younger adults, yet their impression ratings for those faces changed more positively than younger adults'. These findings suggest that older adults may misremember past attributes as positive, regardless of actual valence.

The population of adults over the age of 65 in the United States is growing and is projected to make up over 20 percent of the national population by 2030 (Ortman et al., 2014). Furthermore, as we age, chances of developing age-related disorders—such as dementia—increase (see "2022 Alzheimer's Disease Facts and Figures," 2022 for a detailed report of these trends). With an increasing proportion of adults over 65, studies which examine differences in cognition between older and younger adults (Dennis et al., 2020; Dennis & Cabeza, 2011; Overman & Becker, 2009; Overman & Stephens, 2013; Spreng et al., 2010) have become increasingly important in identifying under what conditions changes in cognition take place, and further, how we might address age-related disruptions to cognition. These changes increasingly impact daily functioning of older adults, impacting things such as the ability to remember appointments, medication dosages and schedules, and due to the increasing impact of these effects on the population of the US—even during the course of healthy aging—the corpus of work examining differences between younger and older adults' cognition has rapidly grown to address these decrements. One specific focus has been on the differences in *memory* performance between younger and older adults (see Old & Naveh-Benjamin, 2008 for a comprehensive review of these differences). Research has shown older adults experience memory deficits related specifically to episodic memory (i.e., memory for time and place details of events; Clark et al., 2018). Researchers have presented a number of hypotheses to explain these observed impairments, both on the "front-end" during encoding; such impairments would be driven by a decline in processing speed, a reduction of processing resources, age-related inhibitory deficits, or decreases in cognitive control capacities (see Kester et al., 2002 for a comprehensive account of these theories), or on the "back-end" during retrieval, where retroactive or proactive interference disrupts recall (see Ebert & Anderson, 2009 for an analysis comparing healthy and

pathological populations). Substantial evidence suggests these declines can be explained in part as an impairment in associative memory ability specifically (Chalfonte & Johnson, 1996; Old & Naveh-Benjamin, 2008a) offers a meta-analysis). Associative memory tasks require participants to link two otherwise unrelated pieces of information together, a capacity which has been shown to diminish as we age (Naveh-Benjamin, 2000; Naveh-Benjamin & Mayr, 2018 for a review of age-related differences in associative memory). Naveh-Benjamin (2000) specifically connects episodic associative memory deficits observed in older adults to a reduced ability to bind unrelated information together formulated as an Associative Deficit Hypothesis (ADH), and demonstrates memory deficits result from older adults' reduced ability to bind unrelated information into a cohesive whole.

In an experiment using words, nonwords, and word-nonword pairs, Naveh-Benjamin (2000) found older and younger adults recalled previously presented words similarly well, yet older adults performed poorer than younger adults when recalling nonwords, and recall rates for word-nonword pairs was even lower than recall for single items. These results indicate older adults can store word information about as effectively as their younger counterparts, yet their ability to store nonword information—and more importantly for this investigation, form associative links between pieces of information (i.e., paired items)—is impaired. Since 2000, numerous studies have demonstrated this associative deficit in episodic memory (Bender et al., 2010; Naveh-Benjamin et al., 2007). While the ADH was originally posited via a word association paradigm, work by Overman & Becker (2009) extends these findings in a word-face paradigm where authors found that while repeated presentations of single items improved memory in both age groups, only younger adults benefitted from repeated presentations of paired items. Further work (Overman et al., 2018, 2019) indicates the configuration in which face-scene

information is presented changes associative memory performance in older adults, suggesting associative deficits can be ameliorated by strategically manipulating external factors, such as contextual information and recapitulation of conditions at study during retrieval. These findings from the field of memory and aging highlight the importance of binding information, the effect that impaired associative memory capacities have on the ability to successfully recall information, and possibilities for improving associative memory in older adults.

As referenced earlier, memory deficits, specifically episodic associative memory deficits, have implications beyond the lab setting. Social interactions, crucially dependent on episodic memory, are particularly affected by deficits to associative memory. For example, remembering who said something and the context surrounding that instance relies on associative links being formed over the course of an interaction. In order to successfully navigate the social world, our ability to bind disparate pieces of information related to people and later recall this information is imperative. Impressions, which are the summation of the pieces of information we perceive, encode, and then bind into bodies of prior knowledge from previous experience and knowledge (Asch, 1946), are an important example of how associative memory is critical for social interactions in the world. We know that, once formed, impressions are not static (Putnam et al., 2014), so it is important that when information about others changes, we are able to encode association changes to update our impressions.

The history of social cognition concerning issues of impression formation and updating have been varied. Uleman & Kressel (2013) provide a detailed account of the history of impression research where they explain that the development of theories of impression formation stem from attempts to explain personality. These endeavors began with the work of F. H. Allport (1924) with his textbook *Social Psychology,* where empirical studies of personality were

outlined. The development of methods to measure subjective phenomena, such as Likert's scale (1932), was an important next step which allowed the study of subjective phenomena such as personality and emotion through a more rigorous, quantitative methodology than self-reporting previously allowed. A focus on attitudes was added to the field with Murphy and colleagues' revised edition of *Experimental Social Psychology* (1937). These developments in psychometric techniques and the shift to a focus on attitudes are the foundation social cognition would later build on to analyze the phenomena of impression formation.

Asch's (1946) seminal work extended the empirical investigation of impressions of others' personality. In his study, participants were asked to form impressions of fictional people presented to them and provide these impressions to the researchers. Results from this work elucidated the role primacy (first impressions are more often recalled than serially subsequent impressions), recency (final impressions also tend to have a higher rate of recall), play in impression formation. It also grants insight into the impact impressions have on our other cognitions about people (i.e., the halo effect explains why the impressions we form, like how nice someone is, impacts our belief about their intelligence as well). To this point, researchers had devoted much attention to studying both personality and memory, and many have produced theories which attempted to explain how impressions are formed and change (again, Asch, 1946 provides a detailed account of these theories). While prolific, these theories lacked the ability to explain important components of impressions formation. The introduction of cognitive psychology to the social psychological domain was the next major advancement for this field, and allowed for the explanation of intricacies of impression formation which had, until this point, been unresolvable. This inclusion allowed for a greater understanding of how mechanisms such as associative memory play a role in impression formation.

The work of Hastie and Kumar (1979), extended by Srull and Wyer (1979), which resulted in an associative network model for how impressions are formed and recalled, proved an important step in the integration of cognitive theory and impression updating. This node-based model hypothesized impressions form as parts of hierarchical nodes. People are at the top of nodes, then traits and finally behaviors are organized in connecting points below. Recalling an impression of someone requires the activation of their node, and subsequent recall of traits and behaviors is represented by activation of these connections of the nodal network. Though this model provides an important theoretical perspective in the formation impressions, it is where application of associative memory concepts to impression updating has ended. The associative network model approaches an explanation by which impression information is retrieved, but it falls short of being able to explain the intricacies of associative memory, such as proactive and retroactive effects, which must necessarily be explained for in an account of the associative memory mechanisms of impression updating is to be considered complete. The latest work in this field (Mende-Siedlecki, 2018; Mende-Siedlecki et al., 2013) introduces neuroimaging techniques to the exploration of associative links and impression updating to define the neural basis of impression updating.

While past work has fallen short, applying associative memory theories to impression updating does have the potential to clarify the specific memory mechanisms by which impression information is stored and recalled. At their most basic level, impressions require the binding of information to the source that information applies to. For example, Putnam and colleagues (2014) used policy position changes of political candidates amid a campaign to examine the effects of detecting change on recall for a candidate's most recent position on an issue. In such a paradigm, accurate impressions must necessarily change as a candidate's

position changes, which requires noticing changes when candidates make them. Thus, impression formation of a candidate requires integrating the facts of a political issue, the stance a candidate takes on the issue, and forming an impression of that stance. Impression formation is only the first step, however. One must be able to recall the initial position in the face of potential future deviations. In other words, once formed, updating an initial impression when a candidate changes their position requires noticing this change. Ultimately, impressions of each candidate are important because they inform how an individual votes. These impressions require not only that you know the facts of the political issues of the day, but also the most recent position of each candidate. The successful retrieval of a candidates' position while discriminating this from the position of a rival requires the successful binding of the facts of each issue to the appropriate candidate and later recalling these pieces of information, a task which requires associative links be formed between each candidate and the stance they take. Further, forming an impression of each candidate based on their position requires successfully recalling the candidates' position and measuring this position against your own beliefs and values. Such a process is complex, and researchers have long sought to understand the mechanisms which underlie this process. Given impressions change, it is important to understand the mechanisms of memory that underlie the updating of impressions if we are to have a richer understanding of the dynamics of interpersonal psychology. A wealth of prior research has helped elucidate the mechanisms of memory and the factors which increase the likelihood of successful binding of information (Clark et al., 2018 builds off the relational memory theory proposed by Konkel & Cohen, 2009). While research has been conducted on how impressions form, these theories have yet to be applied in the context of impression updating, where researchers have predominantly sought to explain updating via

associative nodal models (as discussed above in Hastie & Kumar, 1979; Srull & Wyer, 1979) or trait features, such as moral character (Brambilla et al., 2019).

One framework by which we might understand the role of memory in impression updating is change detection, a theory which accounts for how people detect and later remember when changes occur between two pieces of associated information (see Jacoby et al., 2013 for detection's impact on recollection; Putnam et al., 2014 details when failing to detect change impairs later recall, and Wahlheim, 2014 expands on these proactive effects). This memory-for-change account not only takes into consideration associative memory deficits as a result of aging, it provides an account of proactive and retroactive facilitation and interference effects integral to a complete theory of the mechanism of impression updating. The account hinges on the detection of changes between old and new information, and hypothesizes that when an individual is reminded of old information when they are presented new information, this new information episode serves as a reminding episode, and becomes integrated into a single memory trace. In essence, the old information, new information, and being reminded of the old by the new become integrated into a unitized memory. In the process of impression updating, changes between previously stored and new information must be detected and later recalled for updating to occur successfully. Indeed, Jacoby and colleagues (2013) find a high correlation between the detection of change and later recall, such that when change is detected the ability for an individual to later recollect that change is enhanced. For example, when a change is detected between a list 1 presentation of an associative pair (e.g., fish-pencil) and a list 2 presentation (e.g., fish-balloon,) greater rates of recall result when the change from pencil to balloon is noticed and later recalled. Other work (Jacoby et al., 2015; Jacoby & Wahlheim, 2013) extends these findings and demonstrates the facilitation and interference effects different looking-back-to-notice-change

instructions have on recall (but see Wahlheim, 2014; Wahlheim et al., 2014 for more complete accounts of these effects). To date, however, no investigation has applied change detection theories to impression updating.

Examining the relationship of change detection in impression updating is a natural extension of the work already conducted by social cognition researchers who have long tried to explain how impressions are formed. The current study applies a novel approach to explaining the mechanisms of impression updating by focusing on change detection. To date, impression updating research has not considered change detection and its possible role in associative memory performance related to impression updating. Prior work tells us that impressions are the sum of information we successfully bind and recall throughout our interactions with others (Asch, 1946). Furthermore, we know these impressions are not static (Putnam et al., 2014). Given that impressions change, the importance of understanding the mechanisms that underlie the updating of impressions in memory cannot be overstated. A wealth of prior research has helped to elucidate the workings of memory and the factors which increase the likelihood of successful information binding. However, these theories have yet to be applied in the context of impression updating, which has previously been viewed mainly through a social psychological lens. One framework by which we may understand the role of memory in impression updating is change detection, a theory which accounts for how people detect, and later remember, when changes happen between two pieces of associated information (as before, see Jacoby & Wahlheim, 2013; Jacoby et al., 2015; Wahlheim, 2014; Wahlheim & Jacoby, 2013 for a more thorough account of change detection and recollection). The purpose of the current research is to understand the role change detection, and later recollection, plays in impression updating. The relationship between the two is analyzed in the context of younger and older adults, and seeks to

understand how the process of impression updating might be accounted for by the memory-for-change framework Wahlheim, 2014 proposes.

The current study bridges the fields of cognitive and social psychology by specifically evaluating the theory of change detection as a way of explaining the process of impression formation and updating. The current research will examine how memory processes contribute to forming and updating impressions of other people and attempt to clarify how younger and older adults may use their memory systems differently to accomplish this important social task. Currently there are social psychology explanations for how impressions are formed and updated and there are memory explanations for how information is updated. However, theories from these two different lines of research have not been combined at this point. Our findings inform the understanding of how impressions of others are changed as we learn about them via the process of impression updating. Thus, we sought to understand the role change detection, and later recollection, plays in impression updating, and how impression updating in younger and older adults as a result of changes in personality trait adjectives that are associated with faces might be accounted for by the memory-for-change framework (Wahlheim, 2014).

Method

Participants

41 Older adults (30 women, 11 men; M_{age} = 75.56 years, range 65-91years; $M_{education}$ = 16.49 years, range 12-24 years) were recruited through a community mailing (sent by referencing publicly available state voting records) and an existing lab database of older adults

over the age of 65 in Alamance and surrounding counties. 13 older adults were excluded (8 due to low Montreal Cognitive Assessment—MoCA—scores, and 5 due to technical issues or non-compliance). Summary statistics of included older adults were as follows: 28 Older adults (22 women, 6 men; M_{age} = 74.25 years, range 65-86years; $M_{education}$ = 16.52 years, range 12-24 years). Older adult participants were compensated via entrance into a raffle to win one of five gift cards to a local store and were offered Dana Foundation materials on brain health and aging. A total of 36 younger adults (31 women, 5 men; M_{age} = 18.57 years, range 18-21 years; $M_{education}$ = 13.44 years, range 12–16 years) were recruited from Elon University psychology courses using the Sona Psychology Department participant pool (IRB#11-050). 3 younger adults were excluded due to technical issues or non-compliance. Summary statistics of included younger adults were as follows: 28 women, 5 men; M_{age} = 19.67 years, range 18-21 years; $M_{education}$ = 13.49 years, range 12–16 years. Elon University students received research participation credit or extra credit in their course at the discretion of their professor.

All procedures were approved by the Institutional Review Board of Elon University. All participants were native English speakers. Additionally, older adults provided their health history and all reported no history of major medical, neurological, or psychiatric disorders. Prior to starting the study, all participants read and provided their written consent using an IRB-approved informed consent form and answered a COVID-19 health screening in adherence to Elon University standards for ethical research. Following their completion of the study and cognitive assessment, all participants were debriefed both verbally and in written form. All participants were sent a copy of the debriefing for their records.

Materials

The COVID-19 pre-screening as well as the informed consent form were administered through Qualtrics (Qualtrics, Provo, UT). The study was programmed in E-Prime 3.0 software (Psychology Software Tools, Sharpsburg, PA). Older adult participants completed the Montreal Cognitive Assessment (MoCA; Nasreddine et al., 2005) to measure cognitive functioning (M = 27.57, SD = 1.28), upon completion of the computerized task. 8 older adults who scored below a 26 on the instrument (which indicates possible cognitive impairment) were excluded from final analyses. 132 face stimuli (18-88 years old) developed by Minear and colleagues (Minear & Park, 2004) and 90 valenced adjective stimuli (45 positive-valenced, 45 negative-valenced) developed by Gruhn (2016) were presented to participants. Adobe Photoshop (Photoshop, Ann Arbor, Michigan) was used to remove backgrounds of facial stimuli, as well as memorable visual features of the faces, such as birth marks, scars, moles, and piercings.

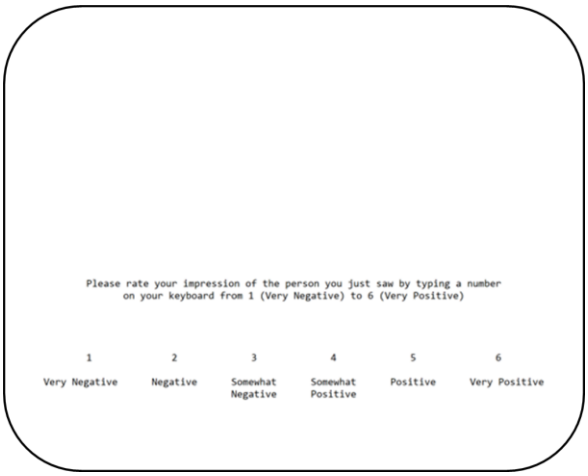
Design

This study used a face-adjective paired-associates paradigm with three phases to measure differences in change detection, recollection, and impression updating of younger and older adults. Phase 1 consisted of an initial impression/impression formation task, phase 2 asked participants to update the impressions formed of faces presented in phase 1, and phase 3, the test phase, measured change recollection via recall of face-adjective associations between phases 1 and 2 in a cued recall test. All instructions and stimuli were presented on a white background with black text to reduce eye strain.

Procedure

After consent and screening, the experiment was presented. Participants were shown a welcome screen which detailed the general layout of the study, and continued to the instructions

for phase 1. In phase 1, participants were directed to study “faces of people at a party [they] had never met before paired with adjectives that describe their personality”. Participants were asked to form impressions of these people based on the adjectives presented with them, and rate their impressions. For the duration of the study, young adult participants were shown young adult faces, while older adults’ participants were shown older adult faces. The researcher remained in the room while the participant read the instructions, and all participants were asked to explain task instructions back to the researcher to ensure their understanding. The researcher clarified any misunderstandings before continuing and left the room before the participant self-initiated phase 1 with a key press once the researcher was out of the room. All instruction slides were succeeded by a fixation cross for 5s before the start of each phase. In phase 1, an impression formation screen with a face-adjective pair was presented (as presented below; faces were not de-identified during study). Participants studied face-adjective pairs for 8s and were then given 8s to rate their overall impression of the face-adjective pair on a Likert rating scale from 1 (very negative) to 6 (very positive).

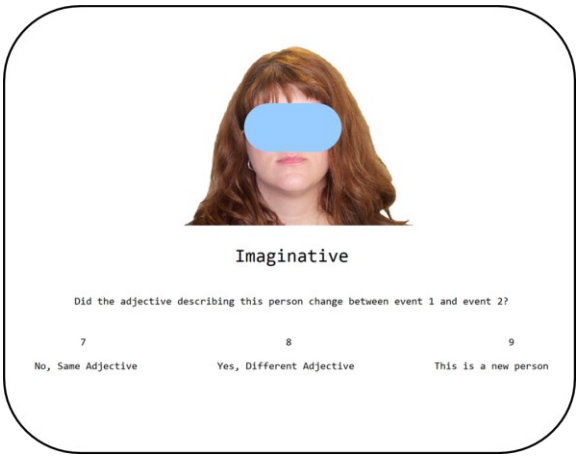


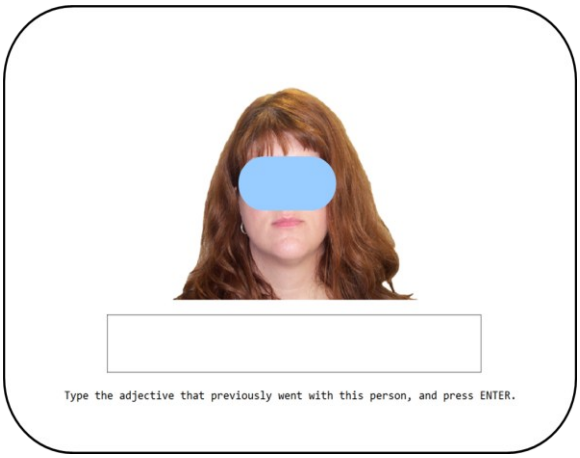
Participants were shown and rated their impression of a randomly paired set of 36 face-adjective associates. Following this, participants were presented with a short answer textbox to describe the strategy used to remember the face adjective pairs they studied. Participants called the researcher back in the room after reading the phase 2 instructions and, like phase 1, were instructed to explain these instructions to the researcher so misunderstandings could be clarified.

In phase 2, participants' ability to detect change and update their impressions were measured. Participants studied intact, changed, and control face-adjective pairs and were asked to recall if the adjective in each pair had changed between phase 1 and 2 (note the changed adjective from the phase 1 diagram).

Of the 48 faces presented in phase 2 six intact pairs were exactly the same as shown in phase 1 (intact pairs); ten were changed, valence-stable pairs (five positive adjectives different from phase 1, five negative adjectives different from phase 1 were presented in phase 2); twenty were changed, valence-changed pairs (ten pairs were positive in phase 1 and became negative in phase 2, and ten were negative in phase 1 and became positive in phase 2); twelve were previously unseen controls (six positively- and six negatively-valenced previously unseen

adjectives paired with previously unseen faces). As each pair was presented, participants indicated whether the face-adjective pair changed from phase 1.

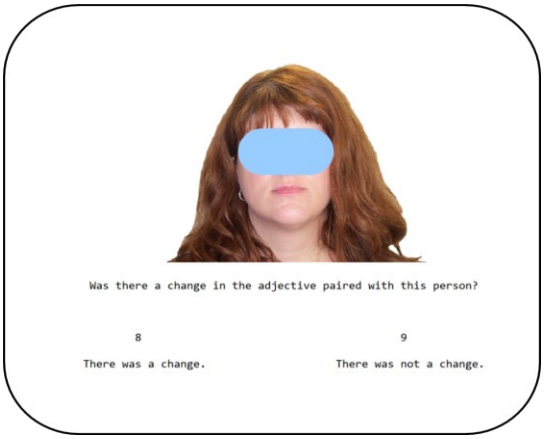




If the adjective was different, they were asked to type the original adjective presented in phase 1. Participants had the opportunity to study stimuli for 8s before they were asked to identify if the adjective describing the face had remained the same, changed, or if the face and adjective had never been seen before. Intact pairs were identified by pressing “7” on the keyboard, changed associates were identified by pressing “8”, and previously unseen controls were identified by pressing “9”. If participants pressed “8” to indicate the adjective associate had changed between phase 1 and 2, they then had 20s to type the adjective they believed was previously paired with the face. Finally, as in phase 1, participants were asked to explain the strategy, if they had one, they used to remember the faces, adjectives, and pairs they studied. Participants then called the researcher back into the room to read and explained the instructions for phase 3.

Phase 3, the test phase, measured participants' change recollection and impression updating. Participants were asked under cued recall conditions to type the adjective paired with each face during phase 2 into a blank textbox. Following this, participants were asked to indicate by pressing "8" if there was a change, or "9" if there was not a change, in the adjective associate of each pair between phases 1 and 2.

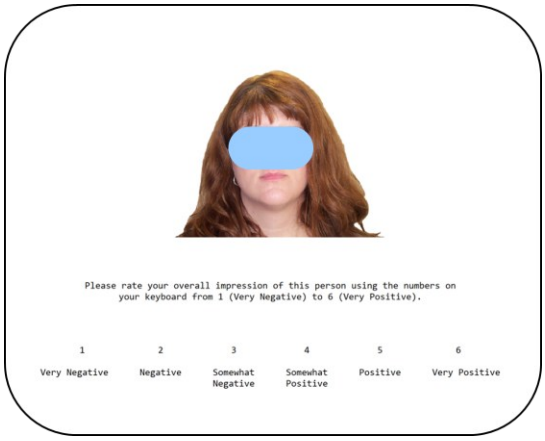
If participants recalled a change occurred, they were prompted to type the adjective associated with that face during phase 1. Next, participants were presented the same Likert rating scale as in phase 1, and were prompted to rate the overall impression of each face. After being presented with all the faces, participants were instructed to call the researcher back into the room, at which time participants were debriefed. Older adult participants also completed the



MoCA assessment at this time to assess their cognitive functioning. After the study, all participants were sent a copy of the debriefing form for their records.

Results

Cued Recall



First, an analysis was run on the correctly recalled adjectives during the phase 3 cued recall task. Figure 1 shows the proportion of correctly recalled adjectives in each pair condition. Correct recall of adjectives was affected by age $F(1,177) = 54.4$, $p < .001$ and pair type $F(2,177) = 36.74$, $p < .001$, and an age by pair type interaction $F(1, 177) = 14.97$, $p < .001$, such that

adjective recall was higher for younger adults than for older adults. Further, while performance improved for both groups when adjectives were repeated, only young adults' performance was substantially improved.

Figure 1. Plot of the proportion of correctly recalled adjectives under cued recall conditions in each face-adjective condition for younger and older adults.

Change Detection

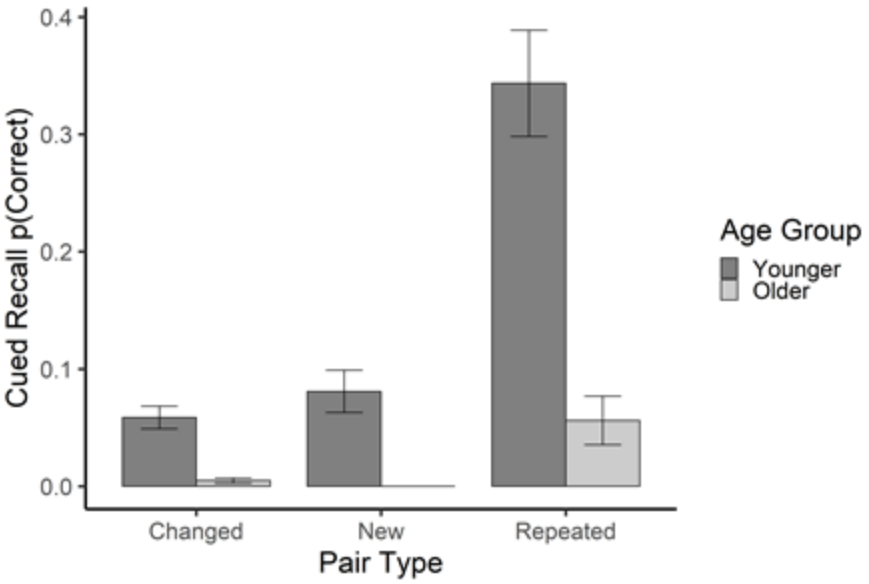


Figure 2.A shows the proportion of correct change detection decisions—measured during phase 2—when participants correctly indicated the adjective of a pair changed between the first and second phase. There was an age effect, such that younger adults performed better than older adults, $F(1,59) = 30.9$, $p < .001$; as well as an effect of pair type, $F(1.8, 106.9) = 15.6$, $p < .001$, though there was no interaction of the two, $F(1.8, 106.9) = 2.08$, $p = .14$ (Greenhouse-Geisser corrected).

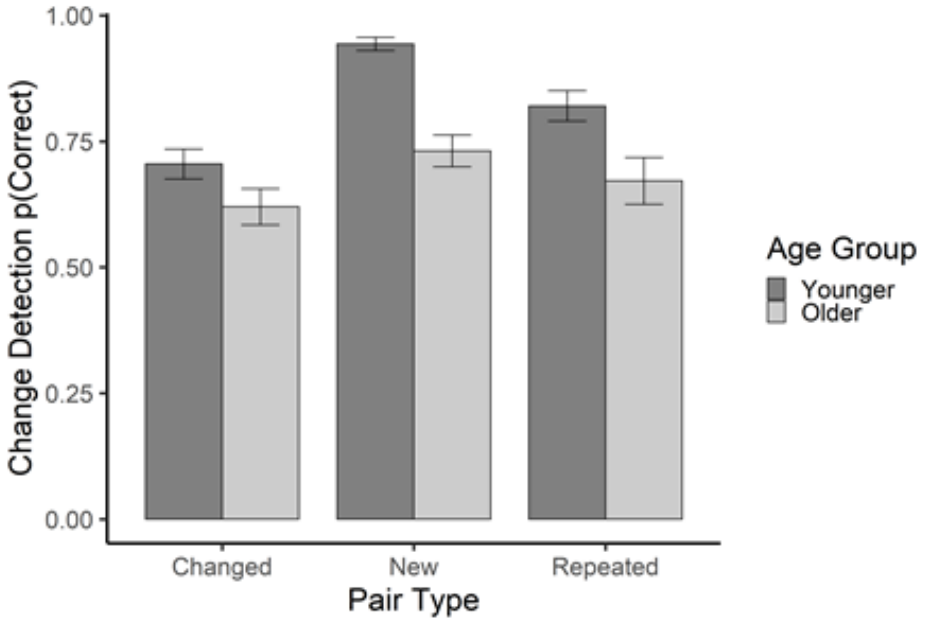


Figure 2.A. Proportion of correct indications of adjective changes by younger and older adults in the three adjective pair type conditions. Younger adults outperform older adults in all pair conditions

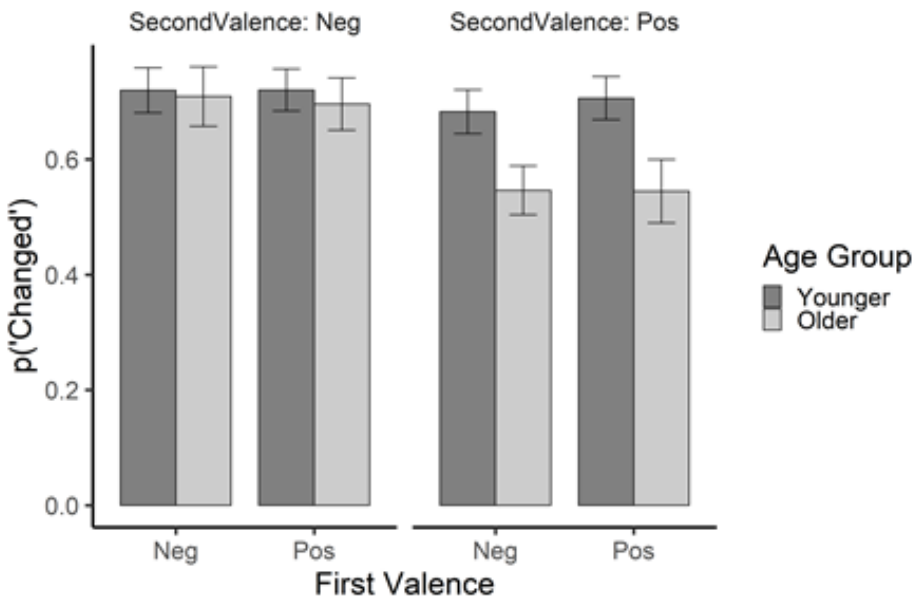


Figure 2.B. Broken down further, valence mediated older adult change detection performance, such that older adults performed as well as younger adults when the second adjective presented to them was negative.

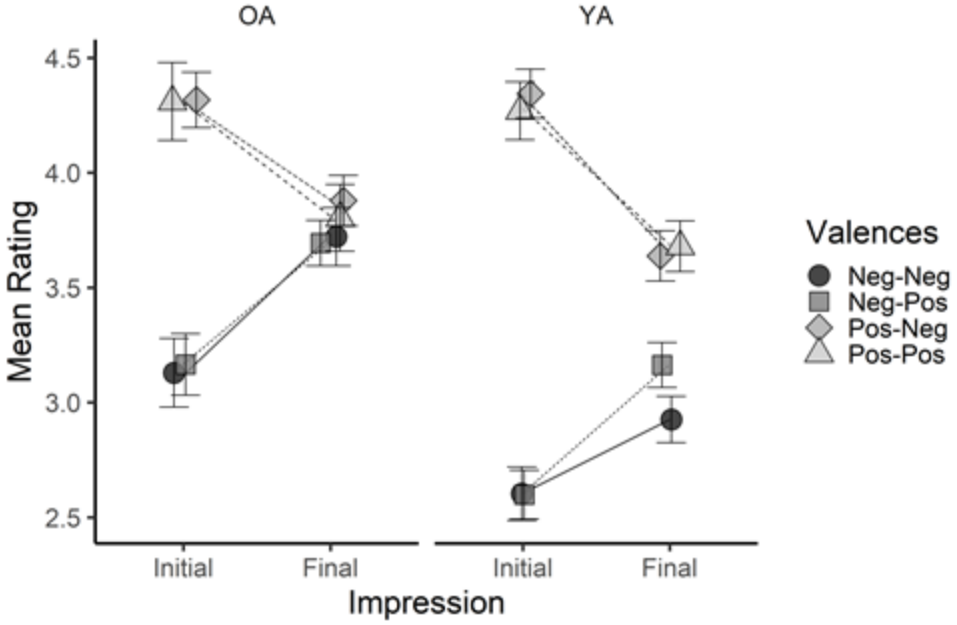
Shown in Figure 2.B, when the effect of pair type was investigated further and the valence of the adjective was accounted for, it was found that older adults' change detection performance was mediated by age $F(1,240) = 7.65$, $p = .01$ and the valence of the second adjective $F(1,240) = 8.21$, $p = .01$, whereby older adults performed as well as younger adults when the valence of the second adjective in the 'changed' pair condition was negative compared to when it was positive.

Impression Updating

Figure 3 shows changes between initial and final impression reports of younger and older adults. Older adults reported more positive average initial impressions than the younger adults. Further, older adults' average impression ratings, even for negative-valenced adjectives, increased. Upon providing final impression ratings, ratings for negatively-valenced adjective

pairs were not significantly different from positively valenced adjective pairs. In contrast, younger adults' average impression ratings were more negative and remained negative compared to the older adults.

Figure 3. Plot of trends between initial and final impressions for younger and older adults. Older adults' impressions were more positive than younger adults. Further, there was no significant difference between impressions for negative first-valence pairs than positive first valence pairs for the older adults.



Discussion

Cued Recall

While performance did improve in both groups under the 'Repeated' condition—though more for younger adults than older adults (validation of findings from Overman & Becker, 2009)—both groups' cued recall was poor. These results are consistent with past work (Craik &

McDowd, 1987) indicating cued recall tasks are more difficult and require more memory resources than other memory tasks, such as recognition tasks. Further, as aging progresses, the memory resources required in cued recall processes decline, compounded the difficulty of such tasks for older adults. Thus, even though both groups did not perform very well, it is not surprising younger adults still outperformed the older adults.

Change Detection

A positivity effect may have been responsible for improved older adult change detection performance when changed adjective associates became negative. Studies (Mather & Carstensen, 2005) have shown older adults have an increased capacity for emotion regulation (i.e., enhanced capacity to protect emotional homeostasis from negative disruption), and this regulation takes the form of preferential attention to, and recall for, positive stimuli as opposed to negative stimuli. Thus, past negative impressions are remembered less than positive impressions because older adults demonstrate preferential attention for positive information, and encode less negative information. This has the effect of casting a positivity bias (Carstensen & Mikels, 2005; Charles et al., 2003; Fernandes et al., 2008; Mather & Carstensen, 2005) over past impressions. The increased salience implied by the contrast between inaccurately positive memories of impressions and new negative impressions may enhance older adult's capacity to notice change.

Though literature in the field supports these conclusions, there may be cause to doubt the strength of these claims. If the results we observed in the current study were caused by a positivity bias as a result of the increased salience between an inaccurate (more positive) impression of an acquaintance and the presentation of negative information incongruent with this previous impression, then we would observe a similar trend of increased detection of change not

only when second adjectives were negative—regardless of the first adjective—but especially when the first adjective was positive and the second adjective was negative. Yet, such a trend does not emerge from these data.

Impression Updating

Past research (Carstensen & Mikels, 2005; Mather & Carstensen, 2005) has documented a positivity effect as aging progresses. Impression updating results from the current study underline this effect. On average, older adults rated their initial impressions of the people they were shown as higher—even when shown a negative-valenced adjective-face pair—than the younger adults. Further, older adults' final impressions were even more positive than their initial impression. These results validate past research, which would suggest this effect is due to a positivity bias found in older adults whereby their memory is better for positive information as opposed to negative information. Thus, older adults' impressions are better for the people they meet, especially their final impressions, because of a memory failure for negative information about these people.

Limitations

Though care was paid to the design and execution of this study, several limitations presented themselves over its course, which, if remedied, would strengthen the results found here and provide guidance to future research. First, this study was started at the onset of the global COVID-19 pandemic. The unprecedented effects of the pandemic on daily life—not to speak of the work of researchers—was dramatic. The sudden transition to quarantine impacted both the physical and cognitive health of people around the globe. Much research has evaluated the impact of stress on cognitive function, including memory performance (Kramer et al., 1991; see

Wolf, 2009 for a review). Given the increased environmental stressors resulting from the pandemic, it is unimaginable memory performance would remain unimpacted. Researchers (Boals & Banks, 2020) are now rapidly attempting to understand the magnitude of the impact on human cognition caused by the Coronavirus pandemic, and just one of the emerging effects the research community has reported anecdotally and is now examining empirically is the memory performance decrements observed across the range of memory research subdisciplines.

There will always be limitations when attempting to use a simulated social experience in a laboratory as a stand-in for a genuine naturalistic one. Studying images of faces on a computer screen to "meet" them is very different from interacting with them in a real-life setting. The decreased salience of features when trying to remember people shown on a screen might have impacted participant's ability to recall adjectives associated with them. This may also have been amplified when taken in the context of prior work (Naveh-Benjamin et al., 2003), which has shown older adults in particular demonstrate associative memory deficits related to recall of picture stimuli. Examining older adult memory always presents unique challenges. Future studies that want to continue to explore our line of inquiry would likely benefit from the use of written vignettes (as has been done in past work by Putnam) or dynamic displays such as video clips (Old & Naveh-Benjamin, 2008b), instead of a static paradigm to measure memory related to social information. Additionally, an experimental design that contains fewer stimuli conditions (such as only positive to negative, or vice versa, adjective changes) with more trials in each of those conditions may reduce task demand while still allow for fine-grained analysis of condition effects.

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